A NOTE ON LIFTING PSEUDOMETRIC ENRICHMENT TO CAUSAL PROCESSES

Elena Di Lavore
- ongoing work with Yoàv Montacute

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Consider a copy-discard category $\mathcal{C}$ with conditionals and suppose that it is (monoidally) $(\mathbf{PMet}_T, +, *)$-enriched, where $* = +$ or $* = \max$.

We want to push this enrichment to the category of causal processes over $\mathcal{C}$, $\mathbf{Proc}_\mathcal{C}$. Suppose that we are given a function $\gamma : [0,T]^{\mathbb{N}} \to [0,T]$. With γ and the enrichment of $\mathcal{C}$, we define a candidate enrichment of $\mathbf{Proc}_\mathcal{C}$:

$$\begin{aligned} d_\gamma : \mathbf{Proc}_\mathcal{C}(A,B)^2 &\longrightarrow [0,T] \\ (f,g) &\longmapsto \gamma(\lambda n.\, d(f_n, g_n)) \end{aligned}$$

We find conditions on γ that make d_γ an enrichment of $\mathbf{Proc}_\mathcal{C}$.

LEMMA 1.
Suppose that

(A1.) γ preserves the pointwise order, i.e.
$$\forall n \; x_n \leq y_n \Rightarrow \gamma(\underline{x}) \leq \gamma(\underline{y})$$

(A2.) γ preserves sums, i.e.
$$\gamma(\underline{x} + \underline{y}) \leq \gamma(\underline{x}) + \gamma(\underline{y})$$

(A3.) γ preserves $(*)$, i.e.
$$\gamma(\underline{x} * \underline{y}) \leq \gamma(\underline{x}) * \gamma(\underline{y})$$

(A4.) γ preserves zero, i.e.
$$\gamma(\underline{0}) = 0$$

Then, d_γ gives a $(\mathbf{PMet}_T, +, *)$-enrichment of $\mathbf{Proc}_\mathcal{C}$.

PROOF

- compositions.
$$\begin{aligned} & d_\gamma(f;g, f';g') \\ &:= \gamma(\lambda n.\, d(f_n;g_n, f'_n;g'_n)) \\ &\leq \gamma(\lambda n.\, d(f_n,f'_n) + d(g_n,g'_n)) && \text{(A1.)} \\ &\leq \gamma(\lambda n.\, d(f_n,f'_n)) + \gamma(\lambda n.\, d(g_n,g'_n)) && \text{(A2.)} \\ &=: d_\gamma(f,f') + d_\gamma(g,g') \end{aligned}$$

- monoidal products.
$$\begin{aligned} & d_\gamma(f \otimes g, f' \otimes g') \\ &:= \gamma(\lambda n.\, d(f_n \otimes g_n, f'_n \otimes g'_n)) \\ &\leq \gamma(\lambda n.\, d(f_n,f'_n) * d(g_n,g'_n)) && \text{(A1.)} \\ &\leq \gamma(\lambda n.\, d(f_n,f'_n)) * \gamma(\lambda n.\, d(g_n,g'_n)) && \text{(A3.)} \\ &=: d_\gamma(f,f') * d_\gamma(g,g') \end{aligned}$$

- zero.
$$\begin{aligned} & d_\gamma(f,f) \\ &:= \gamma(\lambda n.\, d(f_n,f_n)) \\ &= \gamma(\lambda n.\, 0) \\ &= 0 && \text{(A4.)} \end{aligned}$$

- symmetry.
$$\begin{aligned} & d_\gamma(f,g) \\ &:= \gamma(\lambda n.\, d(f_n,g_n)) \\ &= \gamma(\lambda n.\, d(g_n,f_n)) \\ &=: d_\gamma(g,f) \end{aligned}$$

- triangle inequality.
$$\begin{aligned} & d_\gamma(f,h) \\ &:= \gamma(\lambda n.\, d(f_n,h_n)) \\ &\leq \gamma(\lambda n.\, d(f_n,g_n) + d(g_n,h_n)) && \text{(A1.)} \\ &\leq \gamma(\lambda n.\, d(f_n,g_n)) + \gamma(\lambda n.\, d(g_n,h_n)) && \text{(A2.)} \\ &=: d_\gamma(f,g) + d_\gamma(g,h) \end{aligned}$$

□

ex We check that $\sup : [0,T]^{\mathbb{N}} \longrightarrow [0,T]$ satisfies the conditions of LEMMA 1.

(A1.) $\forall n \; x_n \leq y_n \Rightarrow \sup\{x_n \mid n\in\mathbb{N}\} \leq \sup\{y_n \mid n\in\mathbb{N}\}$

(A2.) $\sup\{x_n + y_n \mid n\in\mathbb{N}\}$

$\overset{(A1)}{\leq} \sup\{\sup\{x_n \mid n\in\mathbb{N}\} + \sup\{y_n \mid n\in\mathbb{N}\} \mid n\in\mathbb{N}\}$

$= \sup\{x_n \mid n\in\mathbb{N}\} + \sup\{y_n \mid n\in\mathbb{N}\}$

(A3.) (if $* = +$, then A3 = A2; so, suppose $* = \max$)

$\sup\{\max(x_n, y_n) \mid n\in\mathbb{N}\}$

$\overset{(A1)}{\leq} \sup\{\max(\sup\{x_n \mid n\in\mathbb{N}\}, \sup\{y_n \mid n\in\mathbb{N}\}) \mid n\in\mathbb{N}\}$

$= \max(\sup\{x_n \mid n\in\mathbb{N}\}, \sup\{y_n \mid n\in\mathbb{N}\})$

(A4.) $\sup\{0 \mid n\in\mathbb{N}\} = 0$

□